CSG SOUTH

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QofM

Question of the Month

Will Utility-Scale Battery Storage Ever Be a Reality?

America's power needs are projected to increase 50 percent by 2050. With the growing demand for power comes an increasing need for power production. As states plan for the next quarter-century, policymakers and state leaders must evaluate which battery sources make the most sense from both an economic and operational perspective.

At current prices and on a cost-per-megawatt (MW) basis over the next 25 years, after natural gas, solar energy is the next cheapest option for power generation in America, and some industry experts suggest it may become the overall cheapest if natural gas prices continue to rise at their current rates.

Despite the potential benefit of lower costs, a consistent challenge of solar energy is whether it can ever supply enough power to ensure uninterrupted supply and widespread grid stability. As many in the energy sector are fond of saying, "the sun doesn't always shine" – the implication being that while the sun does not shine around-the-clock, coal and natural gas can always burn, and nuclear fission always spins turbines, maintaining constant supply. Therefore, is solar reliable enough to be part of the solution for the country's growing energy needs? The answer lies in battery storage and the ability to build and maintain enough charge to meet demand during solar's "off hours." This yields the question, will utility-scale battery storage ever be a reality?

The short answer is yes, but with significant considerations surrounding scale, timing, and economics. In 2024, for every 3MW of solar capacity built, 1 MW of storage was added across the country (estimates for 2025 put that ratio at 1.7MW to 1MW). This year, the Electric Reliability Council of Texas (ERCOT), the operator of Texas's power grid, surpassed California as the grid with the most battery storage. California also had forward momentum this year when batteries became the primary source of power for the state's grid between 7:00 p.m. and 9:45 p.m., providing more than 30 percent of power used during that timeframe.

The Battery Storage to Solar Capacity Relationship

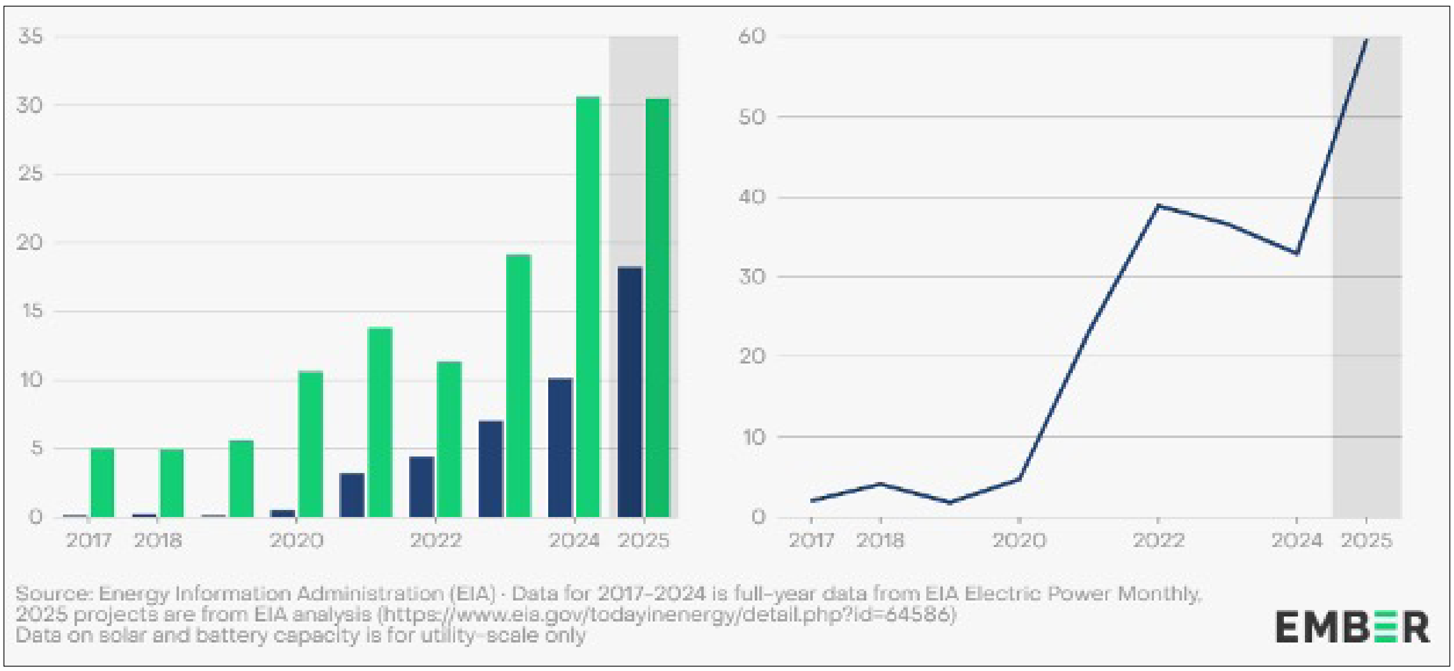
In the US, batteries were a third of solar additions in 2024, expected to reach 60% in 2025

Year-on-year change in capacity (GW)

Battery capacity additions, as a share of solar capacity additions (%)

Solar

Battery

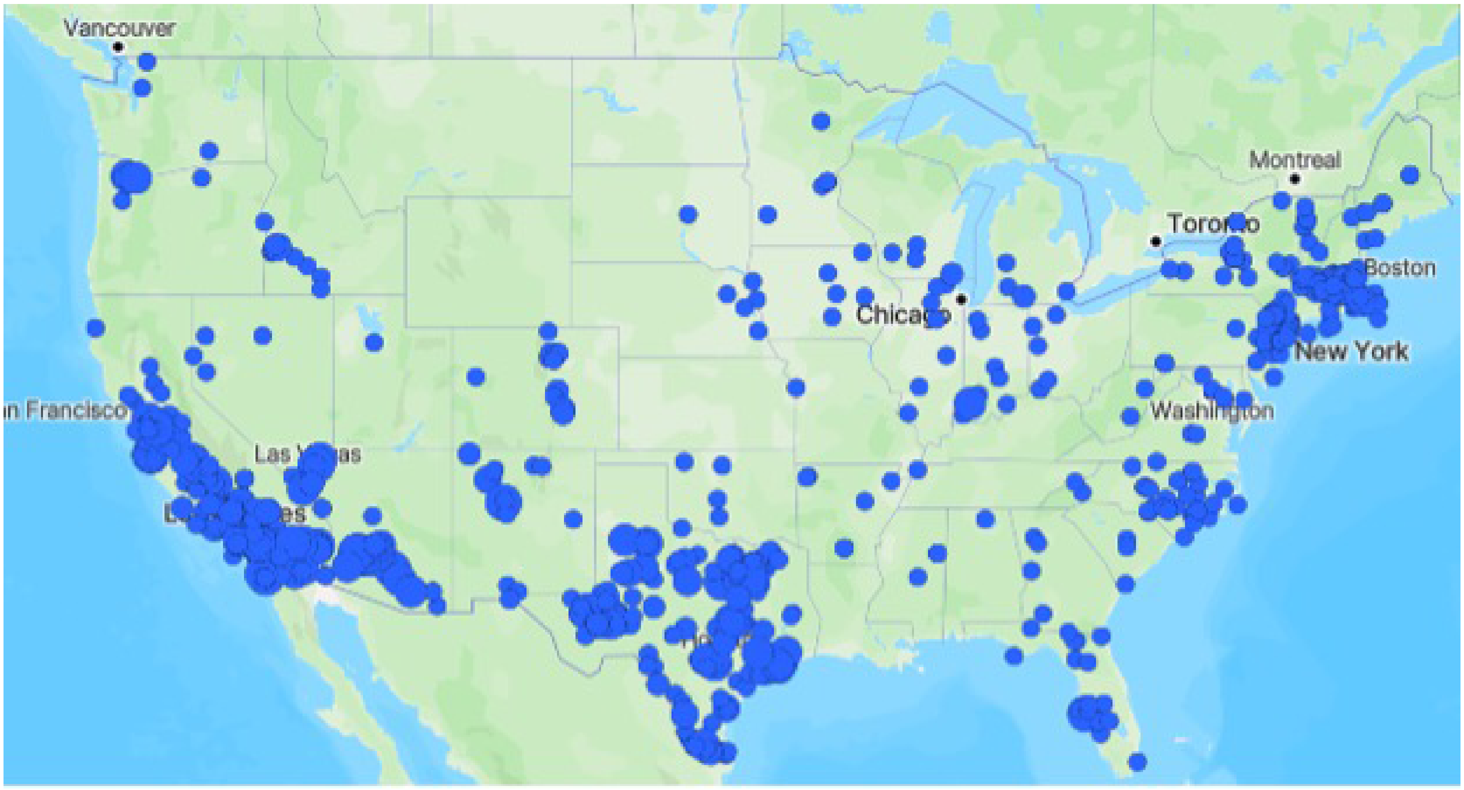


For Southern states, utility-scale battery storage of solar power presents both an opportunity and an infrastructure gap. As of September 2025, every state in the South, except Kentucky and Louisiana, has at least one battery facility with a capacity of 1MW or more. Texas leads the region with 162, followed by North Carolina with 33 and Florida with 17. Across the country, the U.S. has approximately 104GWh of storage online, and the South accounts for just under 20 percent of that amount.

Recognizing the gap, utility providers across the South are expanding facilities and capacity. In Missouri, plans for the Big Hollow Energy Center are underway south of St. Louis in Jefferson County. It will feature 400MW of battery storage to further power the grid. The utility company behind it, Ameren Missouri, plans to bring a total of 1GW of battery storage online by 2030 and 1.8GW by 2042. In South Carolina, Santee Cooper has plans to build a 300MW battery storage system in Berkeley County near Charleston.

Alabama Power has announced plans to build a 150MW battery energy storage system west of Birmingham in Walker County on the site of the former Gorgas coal plant. Expected to be completed in 2027, the system will draw power directly from the grid but will only store enough energy for two hours of operation. Georgia Power is building a similar facility, a 200MW facility southeast of Macon, which is expected to store enough energy for a four-hour period.

[Battery Storage Across the U.S. as of October 2025](#)



The projects in Georgia and Alabama, while indicators of innovation and continuing efforts toward integration of solar energy into providers' production profile, demonstrate the significant amount of battery storage required for only relatively short windows – two to four hours – of power. Because of the rapid growth of energy demand and solar power generation, increasingly efficient solar cell development, and higher-capacity battery technology, there is no clear-cut number or static measure of how much battery storage is needed nationwide to ensure long-term service provision and grid stability. However, the Solar Energy Industries Association (SEIA) has called for 560GWh of grid-scale storage by 2030 to provide adequate nationwide capacity for full grid coverage. This five-fold increase in current capacity would allow batteries to compensate for solar downtime (night) without relying on other energy sources.

While more utility-scale battery facilities are being built, not enough are currently planned to meet SEIA's goal. However, SEIA's estimate does at least provide a general roadmap and target of how much storage it would take for battery storage to balance out times when solar is inactive. This highlights the importance of utility-scale energy storage and why it is likely to become a more significant topic of interest to Southern states.

As solar energy plays an increasingly prominent role in America's energy profile, infrastructure investment, a long-term regulatory framework for solar technology and deployment, and continued adaptation to rapidly evolving technology will be important for state policymakers, regulatory agencies, and energy providers. Evidenced by growth, Southern states are responding to this changing dynamic and looking toward long-term energy solutions.